Seventeen certified configurations, fifteen independent bounds for the no-four-coplanar problem in the cube, and what they do not establish

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Abstract

Let $a(n)$ be the largest number of points of $\{0,\ldots,n-1\}^3$ no four of which are coplanar (OEIS A280537). The published data stop at $a(8) = 20$. We exhibit seventeen configurations, each verified independently, certifying

$$a(9) \geq 23,\ a(10) \geq 26,\ a(11) \geq 28,\ a(12) \geq 31,\ a(13) \geq 32,\ a(14) \geq 34,$$
$$a(15) \geq 35,\ a(16) \geq 38,\ a(17) \geq 38,\ a(18) \geq 41,\ a(19) \geq 43,\ a(20) \geq 45,\ a(21) \geq 47,\ a(23) \geq 50,\ a(24) \geq 52,$$

each verified by two programs sharing no code and using structurally different criteria. We state plainly what is *not* established: no upper bound above $n = 8$ is known to us beyond the trivial $a(n) \leq 3n$, and every configuration below is a lower bound only. We also record a bound of our own that these witnesses refuted, and the reason it looked convincing for seven consecutive values before it failed.

1 Reformulation

Four points are coplanar exactly when they lie in a common plane, and a plane carrying at most three lattice points forbids nothing. Hence

$$S \subseteq [n]^3 \text{ is admissible } \iff |S \cap P| \leq 3 \text{ for every plane } P \text{ with } \geq 4 \text{ lattice points.}$$

Two consequences are used throughout. Three collinear points are forbidden, since with any fourth point they are coplanar; and each of the n planes $x = \text{const}$ carries at most three points, whence

Proposition 1. *$a(n) \leq 3n$ for every n.*

This is the only upper bound above $n = 8$ that we can prove.

2 The bounds

n	lower bound	$3n$	gap	quadruples checked
9	23	27	4	8 855
10	26	30	4	14 950
11	28	33	5	20 475
12	31	36	5	31 465
13	32	39	7	35 960
14	34	42	8	46 376
15	35	45	10	52 360
16	38	48	10	73 815
17	38	51	13	73 815
18	41	54	13	101 270
19	43	57	14	123 410
20	45	60	15	148 995
22	47	66	19	178 365
21	47	63	16	178 365
23	50	69	19	230 300
24	52	72	20	270 725

Seventeen configurations, but fifteen independent statements: the rows for $n = 17$ and $n = 22$ no longer carry their own information, since $a(17) \geq a(16) \geq 38$ follows by monotonicity. This has happened three times as the neighbouring row was improved, and it is the reason to read the table as a set of bounds rather than of values — a row is worth something only while it exceeds its left neighbour.

The configurations themselves are in the repository cited below. The gap to Proposition 1 grows with n; whether this reflects the problem or the reach of the search we cannot say, and Section 6 explains why we believe the question is not decidable by search.

3 Verification

Each configuration was checked twice, by programs written independently and testing different statements. The first computes the 3×3 determinant of every quadruple in exact integer arithmetic. The second takes every triple, forms the normal of the plane through it, and tests the scalar product against every remaining point; it also checks that the points are distinct and lie inside the cube, which the first test does not ask. Before each run the verifier is given a deliberately corrupted witness and must reject it; a check that cannot fail confirms nothing. All runs reported zero violations.

4 What is not established

No configuration proves an upper bound. A witness certifies "at least" and can never certify "at most". The values $a(5) = 13$, $a(6) = 16$, $a(7) = 18$, $a(8) = 20$ rest on exhausted search trees; above $n = 8$ we have Proposition 1 and nothing else. In particular the numbers in the table are not claimed to be maxima, and we expect several of them to be improved.

5 A bound of ours that these witnesses refuted

Setting $\log \binom{N}{m}$ equal to the expected number of coplanar quadruples in a random m-subset, computed *as if the constraints were independent*, gives a threshold that landed on exactly $2n+5$

for $n = 5, \ldots, 12$. Against the values then available its slack ran $2, 1, 1, 1, 0, 0$: monotonically to zero, which reads as a bound becoming tight. It is not a bound. The witness for $a(11) \geq 28$ exceeds $2n + 5 = 27$.

The failure has a mechanism. The estimate treats the forbidden quadruples as independent; they are positively correlated, since a set that has already avoided one plane avoids the next more easily. A first-moment threshold therefore *understates*, and we verified the sign on the companion estimate as well: the quadruple-independence threshold lies below the certified lower bound at every n from 5 to 12, with the deficit growing $-1, -2, -3, -3, -4, -4, -5, -4$. What looked like a bracket around the truth was one biased estimator evaluated at two coarsenings, both leaning the same way.

We record the reading error separately, because it is not specific to this problem: a slack that converges to zero does not distinguish a tight estimate from one about to be crossed, and the two cannot be told apart from inside the model.

6 Effort, and why the growth rate is not measurable by search

The ratio of certified bound to n decreases over the range of the table. We do not read this as a property of the problem. Giving each n an equal time budget on one core and recording how often a randomised search reaches the *known* maximum, we measured a success rate of 8/8, 3/8, 2/8, 1/8 at $n = 5, 6, 7, 8$: the effort required grows roughly geometrically. Equal time is therefore not equal effort, and a curve obtained under a fixed budget records the budget. Deciding whether $a(n)/n$ tends to 3 or to something smaller would need a budget growing exponentially in n.

A second contamination is worth naming. In a search whose target doubles as its stopping condition, asking for K and receiving K is not evidence about K. Several values in an earlier draft of this table were produced that way, agreed with a prediction, and were withdrawn when the target was removed and the same program returned more.

7 The symmetric subspace

All configurations above were found by searching the subspace invariant under a cyclic permutation of coordinates. That subspace is enormously enriched in maxima, and enriched more as n grows. In the two-dimensional analogue, where we hold complete material — every maximum configuration for $n \leq 11$, reproduced against A000755 — the proportion of maxima carrying a symmetry of the square, divided by the same proportion in a null model matched on row and column counts, runs

$$\times 8,\ \times 56,\ \times 125,\ \times 769,\ \times 3729,\ \times 73\,000 \quad (n = 6, \ldots, 11),$$

the last figure resting on seven events in four million and therefore stated to one significant figure only. The raw proportion *falls* across the same range, from 36% to 13%; the ratio to the null model rises.

Enrichment is not containment. At $n = 6$ in three dimensions the cyclically invariant subspace contains no maximum at all: its exhausted maximum is 15 while $a(6) = 16$. Exhaustive traversal of that subspace at $n = 9$ gives exactly 23, which bounds the subspace and says nothing about $a(9)$ beyond the witness.

One consequence deserves emphasis. A sample drawn from a region enriched by three or four orders of magnitude cannot support statements about the structure of solutions in general. Structural claims in this note are made only where complete material is available.

AI/tool-use disclosure

The searches, the programs and the text were produced by autonomous AI agents (Anthropic Claude) under the direction of the author of record, who posed the question, chose what to compute, decided what to claim and takes responsibility for the content. Two agents worked independently: one searched, the other verified and did not read the searcher's code, so that the verification would not inherit its blind spots. Verification protocol, programs, journals and witnesses: `https://github.com/iwasborninbali/saturation`.